

Experiment No. 2

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Experiment Title: Develop a Clothing Store Billing System using HTML, CSS and JavaScript.

Software / Tools Required

- Visual Studio Code
- Google Chrome
- HTML5
- CSS3
- JavaScript (ES6)

Experiment Program Code

Files Used:

- index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Soumya's Clothing Store - Billing System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <header>
      <div class="header-left">
        <h1>Soumya's Clothing Store</h1>
        <p class="subtitle">Professional Billing System</p>
      </div>
      <div class="customer-box">
        <label for="customerName">Customer:</label>
        <input type="text" id="customerName" placeholder="Enter Customer Name" class="cust-input">
      </div>
    </header>

    <div class="main-box">
      <!-- Input Section -->
      <div class="input-side">
        <h3>Add Item</h3>

        <div class="input-grid">
          <div class="form-group">
            <label>Brand</label>
```

```
<select id="brand">
  <option value="">-- Select Brand --</option>
  <option value="Nike">Nike</option>
  <option value="Adidas">Adidas</option>
  <option value="Puma">Puma</option>
  <option value="Levi's">Levi's</option>
  <option value="H&M">H&M</option>
  <option value="Zara">Zara</option>
  <option value="Uniqlo">Uniqlo</option>
  <option value="Other">Other (Type Custom)</option>
</select>
<input type="text" id="custom-brand" placeholder="Enter Custom Brand" style="display:none;
margin-top:5px;">
</div>
```

```
<div class="form-group">
  <label>Item Type</label>
  <select id="name">
    <option value="">-- Select Item --</option>
    <option value="T-Shirt">T-Shirt</option>
    <option value="Jeans">Jeans</option>
    <option value="Shirt">Shirt</option>
    <option value="Shorts">Shorts</option>
    <option value="Jacket">Jacket</option>
    <option value="Sweater">Sweater</option>
    <option value="Dress">Dress</option>
    <option value="Other">Other</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>Size</label>
  <select id="size">
    <option value="">-- Select Size --</option>
    <option value="S">S (Small)</option>
    <option value="M">M (Medium)</option>
    <option value="L">L (Large)</option>
    <option value="XL">XL (Extra Large)</option>
    <option value="XXL">XXL (Double XL)</option>
    <option value="XXXL">XXXL</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>Color</label>
  <select id="color">
    <option value="">-- Select Color --</option>
    <option value="Black">Black</option>
```

```
    <option value="White">White</option>
    <option value="Blue">Blue</option>
    <option value="Red">Red</option>
    <option value="Green">Green</option>
    <option value="Grey">Grey</option>
    <option value="Navy">Navy</option>
    <option value="Beige">Beige</option>
    <option value="Other">Other</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>Price (₹)</label>
  <input type="number" id="price" placeholder="0.00" step="0.01">
</div>
```

```
<div class="form-group">
  <label>Quantity</label>
  <input type="number" id="qty" placeholder="1" min="1" value="1">
</div>
```

```
<div class="form-group">
  <label>GST Rate</label>
  <select id="gst-rate">
    <option value="5">5%</option>
    <option value="12">12%</option>
    <option value="18" selected>18%</option>
    <option value="28">28%</option>
  </select>
</div>
</div>
```

```
<button onclick="addItem()" class="btn-primary">Add to Bill</button>
</div>
```

```
<!-- Bill & Discount Section -->
```

```
<div class="bill-side">
  <h3>Bill Details</h3>
```

```
<!-- Discount Controls -->
```

```
<div class="discount-section">
  <label>Apply Discount</label>
  <div class="discount-controls">
    <input type="number" id="discountPercent" placeholder="%" min="0" max="100" step="0.5">
    <span class="discount-label">%</span>
  </div>
```

```
  <small style="color: #7f8c8d; display: block; margin-top: 5px;">Discount applies to subtotal before
  GST</small>
```

```

</div>

<table>
  <thead>
    <tr>
      <th>Description</th>
      <th>Price</th>
      <th>Qty</th>
      <th>Total</th>
      <th>Action</th>
    </tr>
  </thead>
  <tbody id="list">
    <tr id="empty"><td colspan="5">No items yet</td></tr>
  </tbody>
</table>

<div class="totals">
  <p>Subtotal: <span id="sub">₹0.00</span></p>
  <p class="discount-display" id="discount-display">Discount (0%): <span id="disc-
val">₹0.00</span></p>
  <hr>
  <p>CGST (Half): <span id="cgst">₹0.00</span></p>
  <p>SGST (Half): <span id="sgst">₹0.00</span></p>
  <hr>
  <p class="grand-total"><strong>Grand Total: <span id="final">₹0.00</span></strong></p>
</div>

<div class="action-buttons">
  <button onclick="window.print()" id="printBtn" disabled>Print Bill</button>
  <button onclick="clearBill()" id="clearBtn" style="background-color: #4a5568; margin-top:
10px;">Clear Bill</button>
</div>
</div>
</div>
</div>
</div>
<script src="script.js"></script>
</body>
</html>

```

- [script.js](#)

```

// Global State
let billItems = [];
let discountPercent = 0;

// DOM Elements
const elements = {

```

```

brand: document.getElementById('brand'),
customBrand: document.getElementById('custom-brand'),
name: document.getElementById('name'),
size: document.getElementById('size'),
color: document.getElementById('color'),
price: document.getElementById('price'),
qty: document.getElementById('qty'),
gstRate: document.getElementById('gst-rate'),
list: document.getElementById('list'),
subTotal: document.getElementById('sub'),
discountDisplay: document.getElementById('discount-display'),
discVal: document.getElementById('disc-val'),
cgst: document.getElementById('cgst'),
sgst: document.getElementById('sgst'),
final: document.getElementById('final'),
printBtn: document.getElementById('printBtn'),
emptyRow: document.getElementById('empty'),
customerName: document.getElementById('customerName'),
discountInput: document.getElementById('discountPercent')
};

```

```
// Event Listeners
```

```

elements.brand.addEventListener('change', function() {
  if (this.value === 'Other') {
    elements.customBrand.style.display = 'block';
    elements.customBrand.focus();
  } else {
    elements.customBrand.style.display = 'none';
  }
});

```

```

elements.discountInput.addEventListener('input', function() {
  let val = parseFloat(this.value);
  if (isNaN(val) || val < 0) val = 0;
  if (val > 100) val = 100;
  discountPercent = val;
  updateTotals();
});

```

```
// Add Item
```

```

function addItem() {
  const brand = elements.brand.value;
  const customBrand = elements.customBrand.value.trim();
  const finalBrand = brand === 'Other' ? customBrand : brand;

  const name = elements.name.value;
  const size = elements.size.value;
  const color = elements.color.value;

```

```

const price = parseFloat(elements.price.value);
const qty = parseInt(elements.qty.value);
const gstRate = parseFloat(elements.gstRate.value) / 100;

// Validation
if (!name || !size || !color || !finalBrand || isNaN(price) || price <= 0 || isNaN(qty) || qty < 1) {
  alert('Please fill all fields correctly with valid positive numbers.');
```

```

  return;
}

const item = {
  id: Date.now(),
  brand: finalBrand,
  name,
  size,
  color,
  price,
  qty,
  gstRate,
  total: price * qty
};

billItems.push(item);
renderTable();
updateTotals();
clearInputs();
}

function clearInputs() {
  elements.name.value = "";
  elements.size.value = "";
  elements.color.value = "";
  elements.price.value = "";
  elements.qty.value = '1';
  elements.brand.value = "";
  elements.customBrand.style.display = 'none';
  elements.name.focus();
}

function removeItem(id) {
  billItems = billItems.filter(item => item.id !== id);
  renderTable();
  updateTotals();
}

function clearBill() {
  if (confirm('Are you sure you want to clear the entire bill?')) {
    billItems = [];
  }
}

```

```

    elements.discountInput.value = "";
    discountPercent = 0;
    renderTable();
    updateTotals();
  }
}

function renderTable() {
  elements.list.innerHTML = "";

  if (billItems.length === 0) {
    elements.list.appendChild(elements.emptyRow);
    elements.emptyRow.style.display = 'table-row';
    elements.printBtn.disabled = true;
    return;
  }

  elements.emptyRow.style.display = 'none';
  elements.printBtn.disabled = false;

  billItems.forEach(item => {
    const row = document.createElement('tr');
    row.innerHTML = `
      <td>
        <strong>${item.brand}</strong><br>
        <small>${item.name} ${item.size}, ${item.color}</small>
      </td>
      <td>₹${item.price.toFixed(2)}</td>
      <td>${item.qty}</td>
      <td>₹${item.total.toFixed(2)}</td>
      <td>
        <button class="remove-btn" onclick="removeItem(${item.id})">Remove</button>
      </td>
    `;
    elements.list.appendChild(row);
  });
}

function updateTotals() {
  if (billItems.length === 0) {
    elements.subTotal.textContent = '₹0.00';
    elements.cgst.textContent = '₹0.00';
    elements.sgst.textContent = '₹0.00';
    elements.final.textContent = '₹0.00';
    elements.discountDisplay.style.display = 'none';
    return;
  }
}

```

```

// Calculate Subtotal
const subtotal = billItems.reduce((sum, item) => sum + item.total, 0);
elements.subTotal.textContent = `₹${subtotal.toFixed(2)}`;

// Calculate Discount
const discountAmount = subtotal * (discountPercent / 100);
const discountedSubtotal = subtotal - discountAmount;

// Show/Hide Discount
if (discountPercent > 0) {
  elements.discountDisplay.style.display = 'block';
  elements.discVal.textContent = `₹${discountAmount.toFixed(2)}`;
} else {
  elements.discountDisplay.style.display = 'none';
}

// Calculate GST on Discounted Amount
// Note: GST is calculated on the final price after discount
const totalGst = discountedSubtotal * 0.18; // Assuming avg 18% for simplicity or calculate per item
// Actually, since items have different GST rates, let's calculate per item

let totalCgst = 0;
let totalSgst = 0;

billItems.forEach(item => {
  const itemTotalAfterDiscount = item.total * (1 - (discountPercent / 100));
  // Split discount proportionally is complex, usually discount is applied to total first
  // Simplified: Apply discount to subtotal, then calculate GST on remaining
  // But if items have different GST rates, we should calculate GST per item first, then apply discount?
  // Standard practice: Discount on Item Total, then GST on discounted amount
});

// Re-calculate with per-item GST logic
totalCgst = 0;
totalSgst = 0;

billItems.forEach(item => {
  const itemDiscount = item.total * (discountPercent / 100);
  const itemAfterDiscount = item.total - itemDiscount;
  const itemGst = itemAfterDiscount * item.gstRate;
  totalCgst += (itemGst / 2);
  totalSgst += (itemGst / 2);
});

elements.cgst.textContent = `₹${totalCgst.toFixed(2)}`;
elements.sgst.textContent = `₹${totalSgst.toFixed(2)}`;

const grandTotal = discountedSubtotal + totalCgst + totalSgst;

```

```

elements.final.textContent = `₹${grandTotal.toFixed(2)}`;
}

```

Application Description

The application allows the user to generate a clothing store bill. It supports brand, item type, size, color, price, quantity, GST calculation, discount calculation, printing the bill, and clearing all records.

Features Implemented

Feature	Description
Dynamic Item Entry	Add clothing items to bill
Brand Selection	Predefined and custom brands
Discount	Percentage discount before GST
GST Calculation	Automatic CGST and SGST
Bill Generation	Displays subtotal and grand total
Delete Item	Remove selected item
Print Bill	Print invoice
DOM Manipulation	Updates bill dynamically
Input Validation	Checks valid values

Output

Result / Conclusion

The Clothing Store Billing System was implemented successfully using HTML, CSS, and JavaScript. The application performs dynamic billing, discount calculation, GST calculation, DOM manipulation, and input validation, producing an accurate final bill.